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Unit 2 First Steps with MODI: Unplugged Activities

PDF Documents

PDF Document 1: 1731841790185-Chapter 2_ unit2 First Steps with MODI Unplugged Activities.pdf

This PDF document is included in the unit materials.

YouTube Links

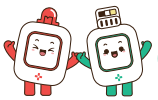
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UNIT 2 FIRST STEPS WITH MODI: UNPLUGGED ACTIVITIES

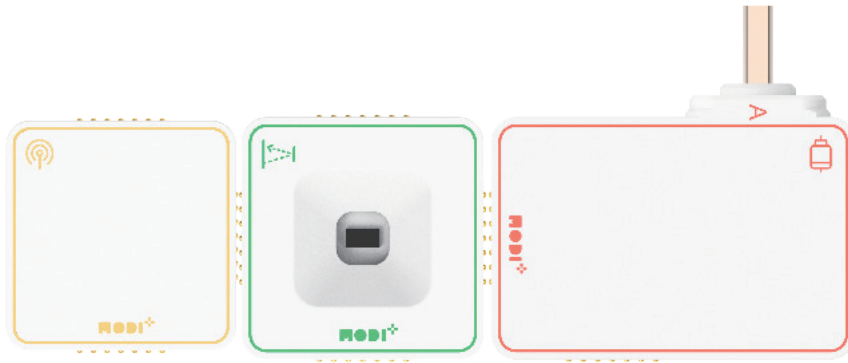
In the world of robotics and programming, an "unplugged activity" is a bit like a hands-on puzzle or game. It doesn't need computers or electronic devices to work.

Before we start playing with our MODI modules, we're going to warm up our brains with some unplugged activities. This will help us think like programmers and understand the steps our MODI robots will need to follow when we start creating with them.



UNPLUGGED

Activity 1 Distance-Sensing Device



Instructions:

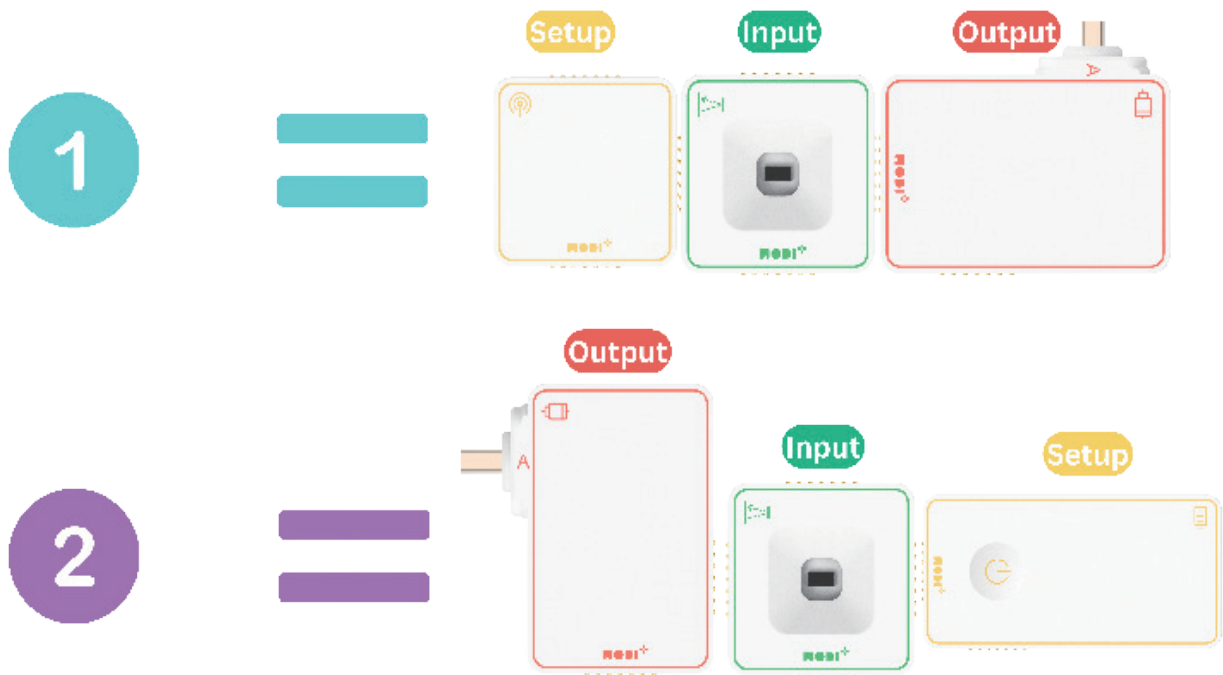
- 1) **Connecting Your MODI Modules:** Just bring your modules close together, and they'll snap into place like magic, thanks to the magnets on their sides.



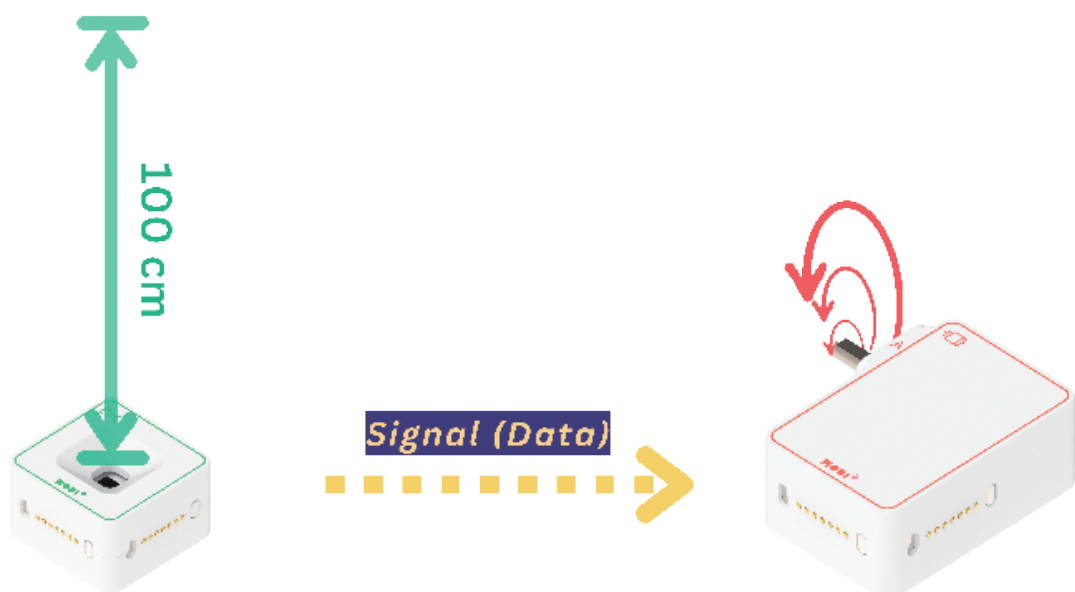


2) **Powering Up Your Modules:** There are two cool ways to make your modules come alive:

- Use the Network module to connect to your PC.
- Use the Battery module for power for the connected modules.



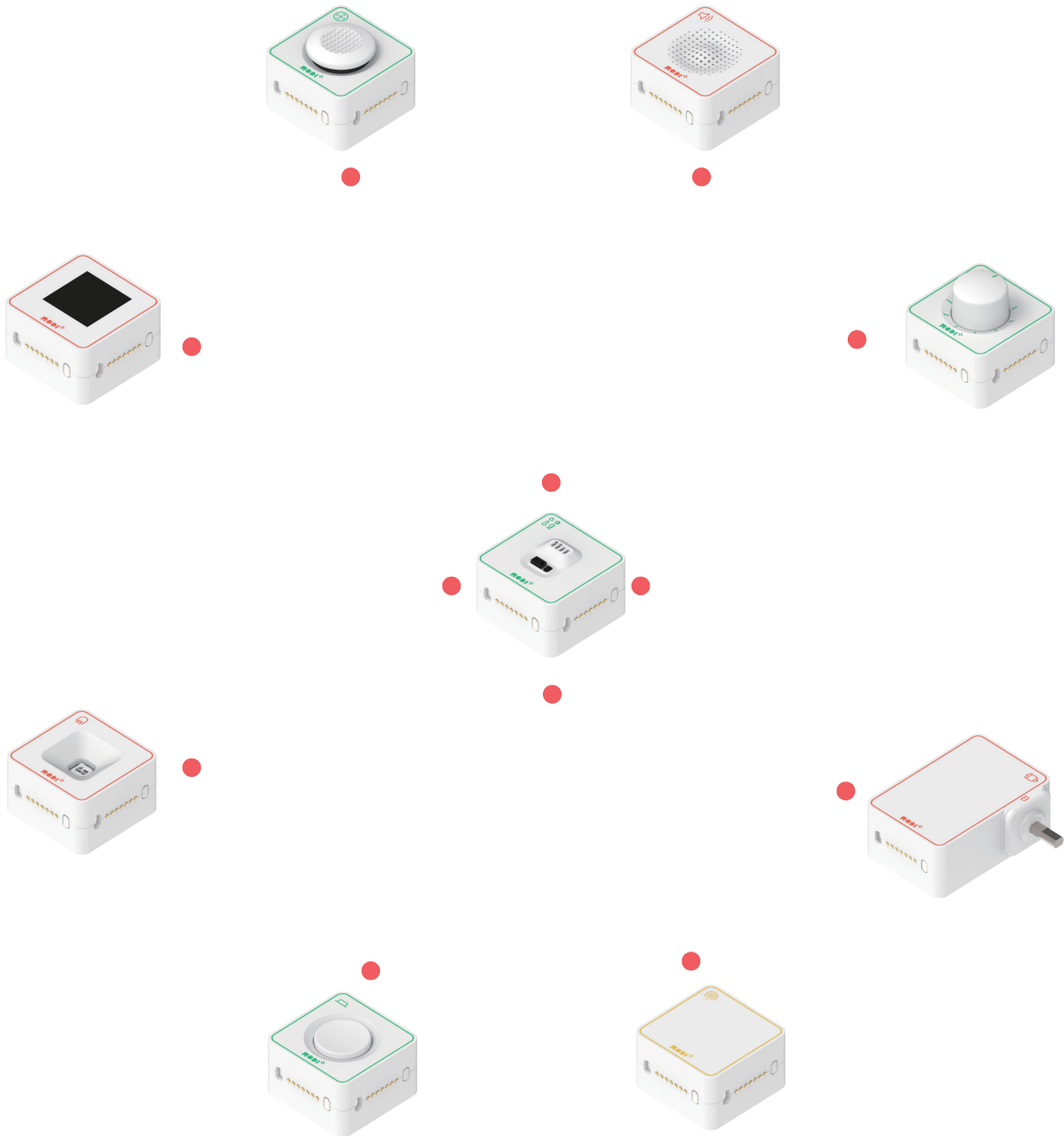
3) **Checking the ToF Module:** The ToF module, short for Time of Flight, is like a tiny radar that can tell how far away things are. It sends out signals and measures how long they take to bounce back, telling us the distance to an object in front of it, up to 100 cm away. When we connect it with a motor module and place it on the floor, it can make some cool things happen. If the path in front of the ToF is clear, it thinks the nearest object is 100 cm away, so it tells the motor to spin really fast. But, if you put your finger right in front of the ToF, blocking its signal, it won't detect any distance. That's when it signals the motor to stop, just like magic! It's a smart way to understand how gadgets like automatic doors work.



Activity 2 Connecting Worlds: Environment and Output

Instructions:

Find 4 MODI modules that can interact with the Environment Module and connect them.



Remember, MODI modules function similarly to components in robotics and computing systems. Ensure your connections between the Environment Module and other modules follow an **"input" to "output" relationship**.



In the blanks below, list the modules you connected to the Environment Module in the "Connecting Worlds" activity. Describe how each pair functions and find similar devices we use every day that work the same way.

Environment +

How it works:

The Environment Module senses changes in its surroundings, such as light or temperature, and based on these changes, it signals Motor A to start moving.

Examples in daily life:

Automatic curtains or blinds operate in our homes, where they open or close in response to sunlight levels detected by a sensor.

Environment +

Environment +

Environment +

